# MAKING 5G HAPPEN

## Understanding the need for 5G and its underlying technologies.

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The buzzword of the masses, 5G, is probably going to be one of the biggest transformational technologies of this decade. It beats the likes of RPA, Blockchain, ZB-era and is right up there with AI at the top. The simple fact is that most of the other technology buzzwords rely on 5G to be the enabling force. AI, for example, requires a massive amount of data bandwidth to derive meaningful information and train neural networks. Most of this is made possible by high-speed interconnects within data centres. However, the moment you have to rely on wireless communication, it slows down to a snail's pace. That's set to change with the deployment of 5G, and not just AI, every major upcoming technology is hoping for a quick deployment to avoid any bottlenecks.

We've been hearing about 2G, 3G, 4G and now there's 5G. What's so special about this generation? Shouldn't it be just another incremental update?

### 5G: THE WORKING

When speaking about wireless communication, we need to understand about radio bands. Every wireless device we use communicates on a fixed frequency band that's supported by it's radio. Your Wi-Fi router works on


2.4 GHz and 5 GHz, if it is dual-band. Or it only runs on 2.4 GHz. Not a lot of devices can communicate on this one band at any given time. So we make use of overlapping frequency bands as individual channels. Thus, the 2.4 GHz could indicate any of the 14 channels which range from 2.412 GHz to 2.484 GHz. Each channel is about 20 MHz wide. And when you have multiple signals crowding a certain channel, you simply move to one that's not so crowded. Moreover, the signal on each channel can incorporate multiple waves by means of using QAM.

That's just one band. 5G will operate on three frequency bands. You have the mmWave which is for spectrum that's greater than 6 GHz, then there's the sub-6 GHz band and lastly you have the sub-1 GHz band. Radio wave frequency has an inverse relationship with the distance that it travels. Lower frequencies travel far and even penetrate multiple layers of concrete in dense urban settings whereas higher frequencies barely penetrate even one layer of concrete and don't travel far. However, you can only get higher bandwidth with high frequencies. 4G LTE operates around 800-1800 MHz so if you want greater bandwidth than 4G, you'll have to switch to the higher frequencies. Doing so presents more challenges. So carriers have the option of picking three different frequency bands based on how they can best serve their